CS & IT ENGINEERING



Algorithms

Analysis of Algorithms



Lecture No.- 07

By- Aditya sir

Recap of Previous Lecture



Topic — Practice of Asymptotic Notations

Topic — Properties

Topics to be Covered



- Topic
- Topic — Properties
- Topic — TC analysis



About Aditya Jain sir

1. Appeared for GATE during BTech and secured AIR 60 in GATE in very first attempt - City topper
2. Represented college as the first Google DSC Ambassador.
3. The only student from the batch to secure an internship at Amazon. (9+ CGPA)
4. Had offer from IIT Bombay and IISc Bangalore to join the Masters program
5. Joined IIT Bombay for my 2 year Masters program, specialization in Data Science
6. Published multiple research papers in well known conferences along with the team
7. Received the prestigious excellence in Research award from IIT Bombay for my Masters thesis
8. Completed my Masters with an overall GPA of 9.36/10
9. Joined Dream11 as a Data Scientist
10. Have mentored 12,000+ students & working professions in field of Data Science and Analytics
11. Have been mentoring & teaching GATE aspirants to secure a great rank in limited time
12. Have got around 27.5K followers on Linkedin where I share my insights and guide students and professionals.



Telegram Link for Aditya Jain sir: https://t.me/AdityaSir_PW



Topic : Analysis of Algorithms

Discrete Properties of Asymptotic Notations:

	Reflexive	Symmetric	Transitive	Transpose Symmetry
O	✓	x	✓	✓
Ω	✓	x	✓	✓
θ	✓	✓	✓	x
o	x	x	✓	✓
ω	x	x	✓	✓

$f = O(g)$
$g = \Omega(f)$

$a < b$
$b < c$
$a < c$

$a < b$
$b > a$



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Symmetry:

If $n = O(n^2)$, then $n^2 = O(n)$ $\Rightarrow$ not true $\longrightarrow$ (not Symm)

If $f = \theta(g)$, then $g = \theta(f)$



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Transitive:

$a \leq b \;\&\; b \leq c \Rightarrow a \leq c$



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Transpose Symmetry:

Real nos.	Asymptotic Notations
a, b	If $f(n) = O(g(n))$,
$a \geq b \Rightarrow b \leq a$?	then $g(n) = \Omega(f(n))$



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Let a, b be two real numbers and f, g be two functions of n.

(1) If $f(n) = O(g(n)) \Rightarrow a \leq b$

(2) If $f(n) = \Omega(g(n)) \Rightarrow a \geq b$

(3) If $f(n) = \theta(g(n)) \Rightarrow a = b$

(4) If $f(n) = o(g(n)) \Rightarrow a < b$

(5) If $f(n) = \omega(g(n)) \Rightarrow a > b$

[MCQ]

#Q. Which one of the following statements is True for all positive functions f(n)?

msg

A $f(n^2) = \theta(f(n)^2)$, when f(n) is a polynomial ✓ A

B $f(n^2) = O(f(n)^2)$ → Small oh

C $f(n^2) = O(f(n)^2)$, when f(n) is an exponential function → Big Oh

D ~~$f(n^2) = \Omega(f(n^2))$~~ $f(n^2) = \Omega(f(n)^2)$ → F

$f(n^2) = O(f(n)^2)$

$(a^m)^n = a^{(m*n)}$

eg:- $f(n) = n^3 \longrightarrow f(n^2) = (n^2)^3$

$= n^{2\times3} = n^6$

$(f(n))^2 = (n^3)^2$

$= n^{3\times2} = n^6$

B) $f(n^2) = o(f(n)^2)$

$f(n) = n$

$f(n^2) = n^2$

$(f(n))^2 = n^2$

$\neq$

c) $f(n^2) = O(f(n)^2)$, $f(n) \rightarrow$ expo (F)

$f(n) = 2^n$

$f(n^2) = 2^{(n^2)}$

$(f(n))^2 = (2^n)^2 = 2^{(2n)}$

2^{n^2} 2^{2n}

$n^2 > 2n$

D) $f(n) = n^2$ $\neq \Omega$

$f(n^2) = (n^2)^2 = n^4$

$(f(n))^2 = (n^2)^2 = n^4$

T

eg2: $f(n) = \log(n)$

$f(n^2) = \log(n^2)$

$= 2 * \log(n)$

$(f(n))^2 = (\log n)^2$

$2\log n \leq \log n \log n$

[MCQ]

#Q. Which of the following is TRUE.

1. f(n) is O(g(n))
2. g(n) is NOT O(f(n))
3. g(n) is O(h(n))
4. h(n) is O(g(n))

} given

[AJ sir Conclusion analysis]

msg

A f(n) is O(h(n)) $f \le h$ → True

B h(n) ≠O(f(n)) $h \not\le f \rightarrow h > f$

$f < g \le h$

C f(n) + h(n) is O(g(n) + h(n))

$f \le g$

$f + k \le g + k$

D f(n).g(n) ≠ O(g(n).h(n)) (F)

$f \ge h$ — $f \not\le h$.

$f \cdot g \not\le g \cdot h$

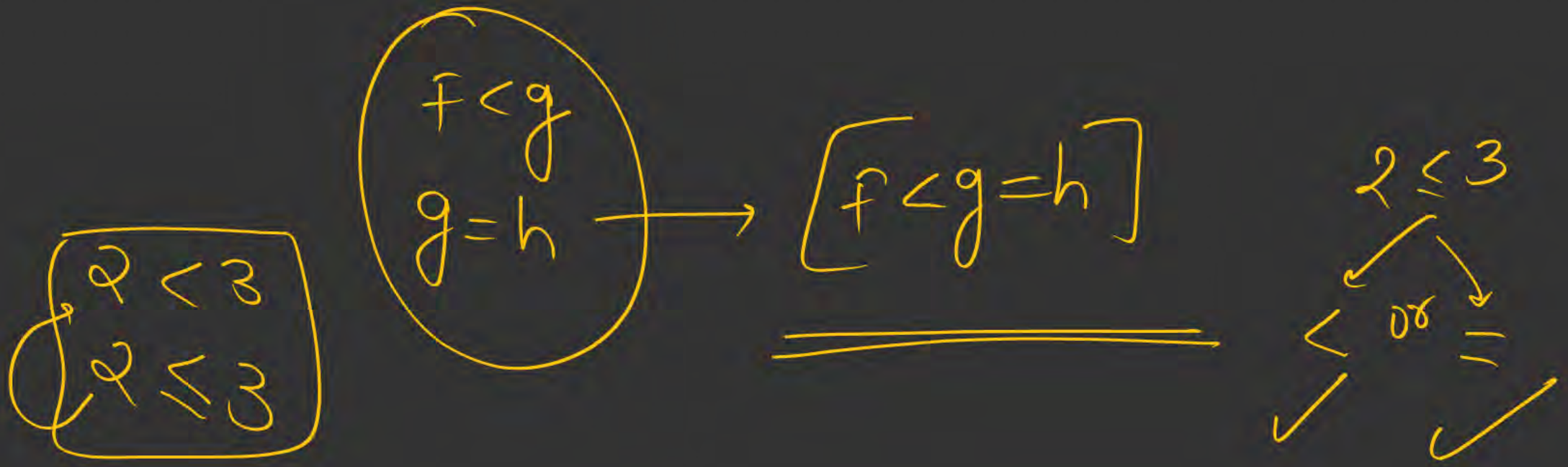
$f(n) = O(g(n)) \longrightarrow f \leq g$

$\not\star$ $g(n) \neq O(f(n)) \longrightarrow g \not\leq f \longrightarrow g > f$ } $f < g$

$g(n) = O(h(n)) \longrightarrow g \leq h$

$h(n) = O(g(n)) \longrightarrow h \leq g$ } $\rightarrow g = h$

$g \geq f$
$g > f$ $\rightarrow g > f$


f < g
g = h
f < g = h
2 < 3
2 ≤ 3
or



Topic : Analysis of Algorithms

(1) Every small oh is also Big Oh, but every Big Oh may or may not be small oh.

(2) Every small omega is also Big Omega, but every Omega may or may not be small omega.

[MSQ]

#Q. $f(n) = n.2^n; g(n) = 4^n$

$f < g$

A $f(n) = O(g(n))$ $f \leq g$

B $f(n) = \theta(g(n))$ $f = g$

C $f(n) = \Omega(g(n))$ $f \geq g$

D None of these

A ∪ C A

$f = n \times 2^n$ $\quad g = 4^n$

$n \times 2^n$ $\quad 4^n$

$(2^2)^n = (2^n)^2$

~~$n \times 2^n$~~ ~~2^n~~ $\times 2^n$

$n < 2^n$

$f < g$

$n \times 2^n$ $\quad$ $4^n = 2^{(2n)}$

$\log_2(n \times 2^n)$ $\quad$ $\log_2(2^{2n})$

$\log n + \log 2^n$ $\quad$ $2n$

~~$n + \log n$~~ $\quad$ ~~$n + n$~~

$\log n$ $\quad$ n

[MCQ]

#Q. Let w(n) and A(n) represent respectively, the worst case and average case running time of an algorithm with input size of n, Which is always TRUE?

MSQ

PYQ

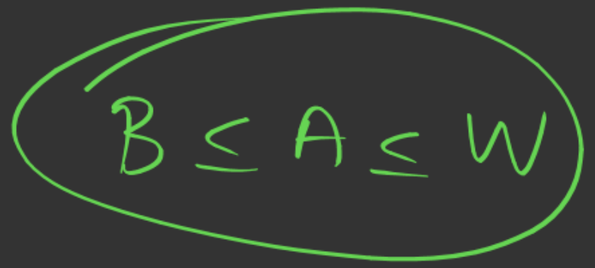
A. $A(n) = o(w(n))$

B. $A(n) = \theta(w(n))$

C. $A(n) = \Omega(w(n))$

D. $A(n) = O(w(n))$

E. $A(n) = \omega(w(n))$


B ≤ A ≤ W

$a \leq b$

$b \geq a$

$$\left[\begin{array}{l} f = O(g) \\ g = \Omega(f) \end{array}\right]$$

$f = n$

$g = n \log n$

$f __ g$

$f < g$

$f = O(g)$

$f = o(g)$

$f = n^2 \log n$

$g = n * (\log n)^{10}$

$n^2 \log n \qquad n * (\log n)^{10}$

$n \times (n/\log n) \qquad (n/\log n)(\log n)^9$

$n > (\log n)^9$

$n > 9 \log_2 n \quad \log n \quad 9 \times \log_2 \log_2 n$

$f = \Omega(g)$

$f = \omega(g)$

Given: $f(n) = n^3, \quad 0 < n < 10{,}000$

$= n, \quad n \geq 10{,}000$

$g(n) = n, \quad 0 < n \leq 100$

$= n^2, \quad n > 100$

msg

$f \geq g$

F A) $f = \Omega(g)$ for $n > 100$

F B) $f(n) = o(g(n))$, for $n > 100$

$f < g$

$f \leq g$

T (c) $f(n) = O(g(n))$, for $n > 100$

T (D) $F(n) = o(g(n))$ for $n > 10{,}000$.

$f < g$

given:

$f(n) = n^3, \quad 0 < n < 10{,}000$

$= n, \quad n \geq 10{,}000$

$g(n) = n, \quad 0 < n \leq 100$

$= n^2, \quad n > 100$

$f(n) = n^3, \quad 0 < n \leq 100$

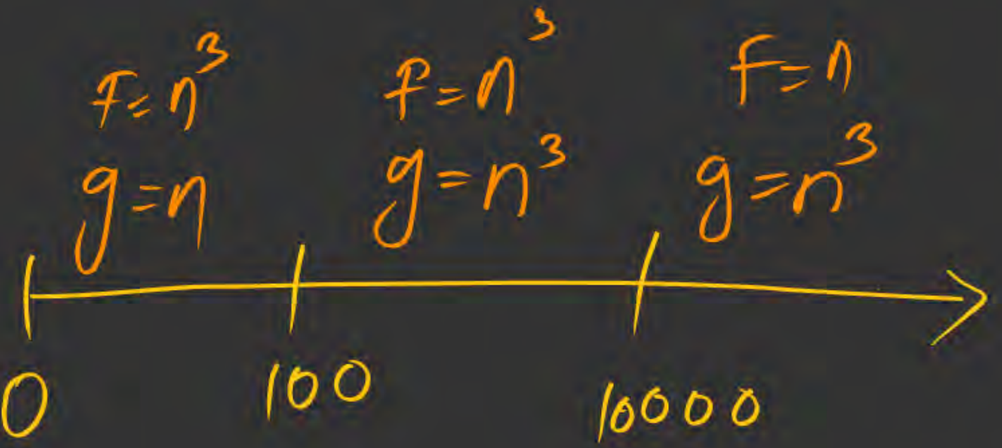
$= n^3, \quad 100 < n < 10{,}000$

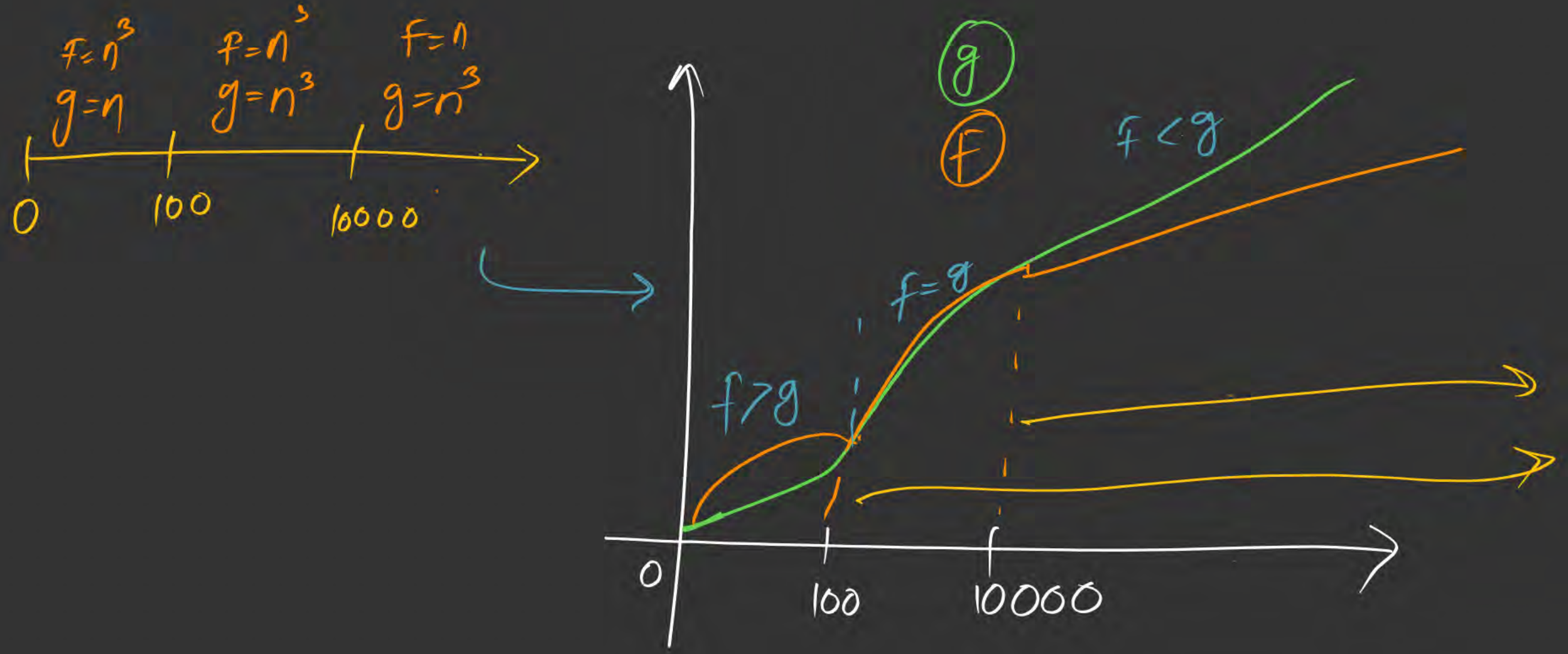
$= n, \quad n \geq 10{,}000$

$g(n) = n, \quad 0 < n \leq 100$

$= n^3, \quad 100 < n < 10000$

$= n^3, \quad n \geq 10000$




f=n^3
g=n
f=n^3
g=n^3
f=1
g=n^3
0
100
10000
g
f
f<g
f=g
f>g
0
100
10000

Asymptotic rate of growth.



$f_1 = 2^n$

$f_2 = n^{1.5}$

$f_3 = n \log n$

$f_4 = n^{(\log n)}$

D

$f_1 < f_4 < f_2 < f_3$

A) f_1, f_4, f_2, f_3 X

B) f_2, f_3, f_1, f_4 X

C) f_2, f_4, f_3, f_1 X

D) f_3, f_2, f_4, f_1

$F_3 < F_2$

$f_1 = 2^n \longrightarrow$ Expo

$f_2 = n^{1.5} \longrightarrow$ Poly

$F_3 = n \log n \longrightarrow$ Poly

$f_4 = n^{\log n} \longrightarrow$ Expo

$f_1 > f_4$

Poly < Expo

$n^{1.5}$

$n \log n$

$n^{1+0.5}$

$= n * n^{0.5}$

$= n\sqrt{n}$

$n * \sqrt{n}$ ✗

$\log n$ ✗

$\sqrt{n} \sim \log n$

2^n $n^{\log n}$

$n > \log n \times \log n$

$n^{1.5}$ $n^{\log n}$

$1.5 \log n < \log n^2$



Summary

→ Practice

→ PYQs



THANK - YOU